

PROJECT JOURNAL

TOPIC: Cross-Language Personalization for Dataset Selection Using BAAI General Embedding

TEAM-MEMBER:

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WEEK 1

Date: 7/12/24

Action: Selecting a data set.

In the first week, I focused on selecting and preparing a dataset to guide the development of a Kannada chatbot. This step is important in constructing a sturdy foundation for the chatbot's average performance. After comparing numerous options, I additionally summarized the Southern dataset for its installed content and adaptability.

Operation:

- Check the dimensions of the dataset to make it well matched with the demanding situations and goals.
- Standardized Kannada text material with help in eliminating differences between Romanized and Native scripts.
- Transliteration mapping identifies and provides important functions such as usage patterns.

Time required: approximately 6 hours

Challenges encountered:

- Handle global script inconsistencies within datasets
- Remove barriers to the availability of great Kannada datasets.
- Make sure that the transliteration is both accurate and meaningful in context.

Solutions used:

- Information is checked by references and manual corrections.
- Improved transliteration compatibility using an iterative pre-processing pipeline.
- Optimized the dataset assignment strategy to reflect Kannada's linguistic nuances.

Lessons learned:

- Improve your knowledge of the complexities of the Kannada language and its challenging transliteration terms.
- Enjoy growing your income to create ready-made datasets for NLP applications.
- Improved data cleaning and standardization skills for multilingual tasks.

WEEK 2

Date: 11/12/24

Task: Data Analysis

During the second week, I focused on studying the dataset to extract linguistic insights and styles essential for tailoring chatbot interactions. Exploratory Data Analysis (EDA) modified into hired to delve into the Kannada language's syntactic and contextual developments.

Actions Taken:

- Conducted phrase frequency analysis to turn out to be aware of normally used phrases and phrases.
- Analyzed metadata together with sentence period, word be counted quantity, and contextual usage patterns.
- Distinguished amongst formal and casual textual content instructions to optimize chatbot responses.

Time Spent: Around 5 hours.

Challenges Faced:

- Differentiating formal and casual texts due to overlapping contexts.
- Handling noisy facts because of uncommon or area-particular terms.
- Accurately representing the linguistic variety of Kannada in a based layout.

Solutions Applied:

- Filtered out low-frequency phrases and beside the point facts to recognition on core styles.
- Created a context-based categorization approach for better text category.
- Refined facts subsets iteratively to obtain clarity and consistency.

Learnings Gained:

- Enhanced know-how of Kannada syntax and usage variations.
- Gained enjoyment in filtering and classifying complicated datasets.
- Strengthened analytical abilities for identifying linguistic trends in multilingual corpora.

WEEK 3

Date: 18/12/24

Task: Create: Embedding

This week I focused on creating an embed to convert Kannada text into machine readable form. Leveraging BAAI General Embeddings (BGE), I ensured that the vector captured the syntax and semantic nuances of the language.

Operation:

- Tokenize Kannada text to prepare for embedding version.
- Use BGE to create high-quality vector representations that retain their linguistic meaning.
- Embeddings are verified by evaluating syntactic similarity and contextual relevance.

Elapsed time: about 12 hours

Challenges encountered:

- Difficulty in creating tokens due to the complex morphology and affiliation of the Kannada language.
- Ensure that the embeddings effectively capture the unique features of the language.
- Fixed performance issues when creating large embeds.

Solutions used:

- Use specialized libraries like INDIC-NLP for efficient tokenization.
- Repeated testing to optimize implantation parameters and improve quality
- Improves the embedding process by leveraging pre-trained models for efficiency.

Lessons learned:

- To develop advanced skills in creating embeds for regional languages.
- Improved to balance semantic depth with computational efficiency.
- Gain insights to evaluate NLP application embeddings

WEEK 4

Date: 27/12/24

Task: Chatbot development and testing

Last week I tested the chatbot integration to ensure smooth communication in Kannada. Special care was taken to handle and transcribe questions into mixed language effectively.

Operation:

- Use a custom language model for Kannada-specific conversation streams.
- Chatbot responses are tested for accuracy, speed, and context awareness.
- Optimize the chatbot's ability to handle a wide range of questions. Contains a mixture of Kannada and English.

Elapsed time: about 11 hours

Challenges encountered:

- Handles transliteration and mixed language questions without errors.
- Balances computational performance with conversational accuracy.
- Refine the chatbot's understanding of contextual nuances in user input.

Solutions used:

- Developed a hybrid system to process mixed language input efficiently.
- Iteratively fine-tune the model. Based on real-world search data
- Improved chatbot evaluation through extensive testing and user feedback.

Lessons learned:

- Expert in developing multilingual chatbots for regional languages.
- Learn how conversational AI can adapt to mixed language environments.
- Get hands-on experience testing and improving chatbot functionality

SUMMARY

Total time spent 34+ hours

Important achievements:

- The Kannada dataset was created as a standard for NLP work.
- Perform detailed data analysis to find important language patterns
- Create high quality embeds based on Kannada language processing.
- Develop and test chatbots that can accurately handle Kannada and mixed language interactions.